

Identifying Interaction Design and Functional Layout

Practical implementation of interaction design and functional layouts

Case study: **Personal App — Calculator, Life Tracker & Settings Modules**

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Tools: Flutter 3.22+ · Dart 3.7+ · Provider · SQLite

1. Aim

To identify user requirements for a calculator, life tracker, and settings management system, design interaction prototypes, and implement them with consistent placement, wording, and colour so the user can build an accurate mental model — demonstrated through the **Calculator, Life Tracker, and Settings modules** of Personal App.

2. Objectives

1. Understand the principles of interaction design and functional layout.
2. Translate user requirements into working design prototypes for Calculator, Life Tracker, and Settings (Activity 1).
3. Apply consistency rules across every screen in all three modules (Activity 2).
4. Evaluate whether the resulting interface supports an accurate, predictable mental model.

3. Theory

3.1 Interaction Design

Interaction design focuses on the sequence of screens and states a user moves through to complete a task. In Calculator, this involves entering expressions, viewing results, toggling scientific mode, and accessing history. In Life Tracker, this includes setting a date of birth, viewing the live dashboard with real-time counters, and adjusting life expectancy. In Settings, this covers theme selection, data management (export/import/clear), and notification toggles.

3.2 Functional Layout

The Calculator uses a structured button grid with a clear display area at the top. The Life Tracker uses a card-based dashboard layout with large numeric readouts. Settings uses a section-based scrollable layout with grouped controls (Appearance, Notifications, Data, About).

3.3 Consistency Principles

- **Placement** — Primary actions are in consistent positions. Settings uses the same dialog patterns as other modules.
- **Wording** — “Save”, “Cancel”, “Delete”, “Discard”, “Retry” use identical labels across all three modules.
- **Colour** — Blue = primary action, red = destructive, amber = warning, green = success.

3.4 Mental Model

When interface patterns stay consistent across Calculator, Life Tracker, and Settings, the user can transfer knowledge from one module to another, reducing cognitive load.

4. Software / Tools Used

Tool	Purpose
Flutter 3.22+	Cross-platform UI framework (Material 3)
Dart 3.7+	Programming language with sound null safety
Provider	State management (CalculatorProvider, LifeProvider, SettingsProvider)
SQLite (sqflite)	Local database (calculator history, settings key-values)
shared_preferences	User preferences (theme, scientific mode, etc.)
share_plus / file_picker	Export/import data via JSON files
package_info_plus	App version display
intl	Date/time formatting

5. Activity 1 — Design and Prototype the Interaction

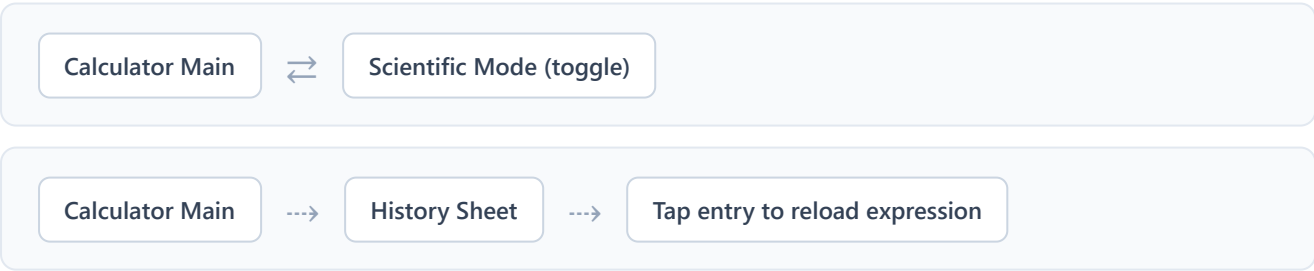
Module 1

Calculator

5.1 User requirements — Calculator

#	User requirement	Screen / interaction
1	Perform basic arithmetic operations.	Button grid — +, −, ×, ÷
2	Use parentheses for expression grouping.	Button grid — (and) buttons
3	Toggle scientific functions when needed.	SCI toggle — adds sin, cos, tan, log, ln, sqrt, π, e
4	Store and recall values with memory.	Memory row — MC, MR, M+, M- with indicator light
5	View and reuse past calculations.	History bottom sheet — last 50 entries, tap to reload
6	Clear or backspace entries.	Clear (C) and backspace (←) buttons
7	Copy result to clipboard.	Optional copy-on-tap setting in preferences
8	See clear error messages on invalid input.	Display shows “Error” for invalid expressions
9	Smart number formatting for readability.	Auto-integer, trims trailing zeros, exponential for >15 chars

Screen flow — Calculator



Views — Calculator

Calculator Module — View Inventory

Category	Views
Modes	Basic Mode, Scientific Mode
History	History Sheet (with entries), History Sheet (empty)
States	Normal display, Error display, Memory active indicator

Module 2

Life Tracker

5.2 User requirements — Life Tracker

#	User requirement	Screen / interaction
1	Set date of birth to begin tracking.	DOB picker dialog — date selector with save
2	Set life expectancy (default 80 years).	Life expectancy dialog — number input 1–120
3	View elapsed time in years/months/days.	Dashboard — elapsed stat block
4	View life progress as percentage with colour coding.	Progress bar — <50% blue, <80% amber, ≥80% deep orange
5	View expected remaining time.	Dashboard — remaining stat block
6	See real-time metrics (days, weeks, hours, minutes, seconds).	Metric cards — auto-updating every second via stream
7	Reset life tracker data.	Reset confirmation dialog
8	Clear guidance when no DOB is set.	Empty state — illustration + “Set your date of birth” CTA

Screen flow — Life Tracker



Views — Life Tracker

Life Tracker Module — View Inventory

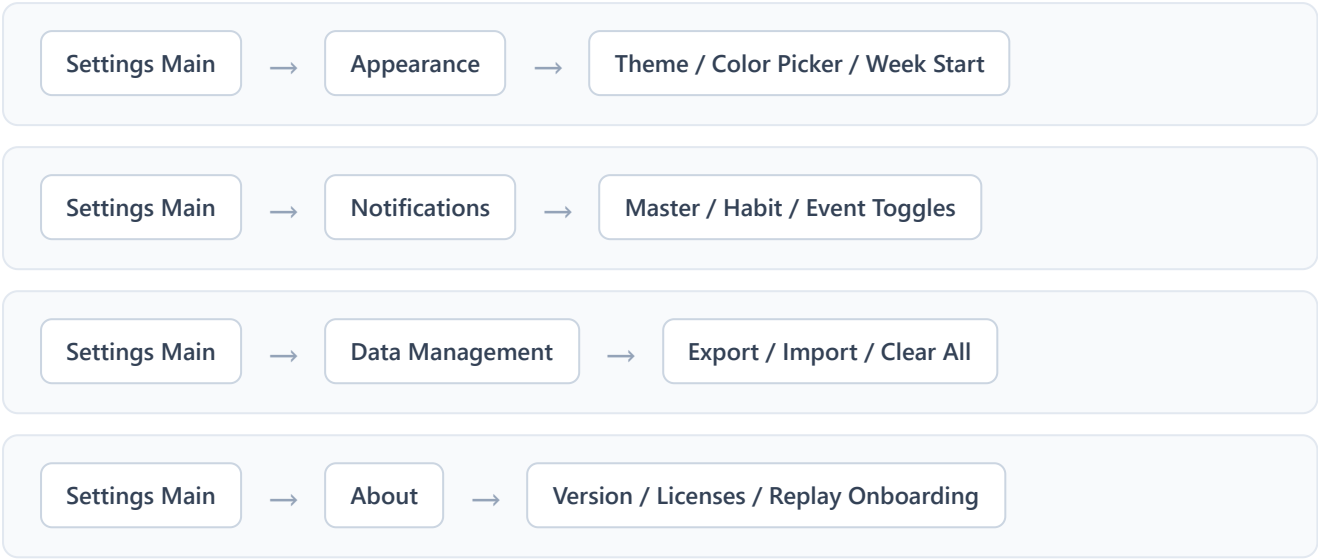
Category	Views
States	Empty State (no DOB), Live Dashboard
Dialogs	DOB Picker, Life Expectancy input, Reset confirmation
Display	Elapsed block, Progress bar, Remaining block, Metric cards (4)

Module 3 Settings

5.3 User requirements — Settings

#	User requirement	Screen / interaction
1	Switch between Light, Dark, and System theme.	Theme toggle — segmented button, persisted immediately
2	Customize accent colour with live preview.	Color picker — 16 presets + custom HSV/Hex with SV gradient + hue slider
3	Choose week start day (Monday vs Sunday).	Toggle switch — affects Calendar module
4	Enable or disable all notifications.	Master toggle — requests runtime permission
5	Toggle habit reminders.	Habit reminder toggle switch
6	Toggle event reminders.	Event reminder toggle switch
7	Export all data to a JSON file.	Export button — generates JSON, opens system share sheet
8	Import data from a JSON backup.	Import button — file picker, validates + DB transaction
9	Clear all app data with two-step confirmation.	Clear button — must type “DELETE” to confirm
10	View app version and open source licenses.	About section — version, GitHub link, licenses page
11	Replay the onboarding tutorial.	Onboarding replay link — resets SharedPreferences flag
12	Configure calculator preferences.	Calculator section — scientific mode, copy-on-tap, clear history

Screen flow — Settings



Views — Settings

Settings Module — View Inventory

Category	Views
Sections	Main Settings (scrollable), Appearance, Notifications, Data Management, About
Pickers	Color Picker (16 presets), Custom Color (HSV/Hex dialog)
Data	Import (file picker), Import (loading), Import (success/error), Clear All (two-step)
Info	About section, Licenses page

6. Activity 2 — Apply and Verify Consistency

Consistency rules were applied once and reused across Calculator, Life Tracker, and Settings via the shared Material 3 design system and common widget library.

6.1 Consistency rules applied

Button Placement

Calculator uses a structured button grid. Life Tracker has a prominent “Set Date of Birth” CTA in its empty state. Settings uses tappable section cards. Destructive actions always require confirmation dialogs.

Label Wording

‘Save’, ‘Cancel’, ‘Delete’, ‘Discard’, ‘Retry’, ‘Apply’ never change. Error messages consistently use ‘Something went wrong’ with Retry. ‘Clear All Data’ uses explicit two-step wording.

Colour Scheme

-  **Blue** — primary action, active tab, equals button
-  **Red** — destructive, clear all data, delete, high priority
-  **Green** — success, import success, low priority
-  **Amber** — warning, medium priority, life progress >80%

Shared Design System

All three modules use the same Material 3 theme with colorSchemeSeed. The bottom NavigationBar persists. Settings changes (theme, colour) instantly apply to Calculator and Life Tracker. Dialog patterns and snackbar feedback are identical.

6.2 Consistency Matrix

Cross-Module Consistency Matrix — Calculator, Life Tracker & Settings

Rule	Calculator	Life Tracker	Settings
Blue = primary action	✓	✓	✓
Red = destructive	N/A	✓	✓
Confirm before delete	N/A	✓	✓
Same wording for actions	✓	✓	✓
Error/Loading/Empty states	✓	✓	✓
Shared theme applies	✓	✓	✓
Navigation persists	✓	✓	✓
Snackbar feedback	✓	✓	✓

7. Output / Result

The Calculator, Life Tracker, and Settings modules are implemented within a single Flutter application sharing the Material 3 design system. The Calculator supports basic and scientific modes with memory, history, and smart formatting. The Life Tracker provides a real-time dashboard with elapsed/remaining time, progress percentage, and live counters that update every second. Settings centralizes theme control, notification preferences, data management (export/import/clear), and calculator configuration. All three modules share the same colour semantics, dialog patterns, and navigation structure.

7.1 Self-evaluation checklist

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- ✓ **Primary action consistently placed?** Yes — Calculator uses predictable grid; Life uses CTA button; Settings uses section cards
-
- ✓ **Same action = same word?** Yes — Save/Cancel/Delete/Discard/Retry/Apply identical across all 3 modules
-
- ✓ **Each colour = one meaning?** Yes — Blue=action, Red=destructive, Green=success, Amber=warning
-
- ✓ **Predict unseen screens?** Yes — Settings reuses dialog patterns from other modules
-
- ✓ **Loading/error states?** Yes — Calculator shows Error; Life has empty state; Settings has import loading/success/error
-
- ✓ **Destructive actions protected?** Yes — Clear All Data requires typing "DELETE"; Reset Life requires confirmation
-
- ✓ **Empty/populated states?** Yes — Life Tracker has empty state; Calculator history has empty state
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- ✓ **Navigation persists?** Yes — 6-tab IndexedStack with NavigationBar/NavigationRail
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Annexure A — Selected UI/UX Ideas Integrated

1. Two-Mode Calculator with Smart Toggle

Module: Calculator | Users toggle between Basic and Scientific mode. Scientific mode adds trigonometric functions (sin, cos, tan), logarithmic (log, ln), square root, and constants (π , e). The toggle state is persisted to SharedPreferences.

2. Memory Functions with Visual Indicator

Module: Calculator | Four memory buttons (MC, MR, M+, M-) store and recall values. A memory indicator light shows when a value is stored, providing clear feedback.

3. Live Real-Time Life Counter

Module: Life Tracker | Uses Stream.periodic to update elapsed time every second. All metrics (days, hours, minutes, seconds) refresh in real time, creating a compelling “clock” effect that encourages daily engagement.

4. Colour-Coded Life Progress Bar

Module: Life Tracker | The progress bar changes colour based on percentage lived: blue for <50%, amber for <80%, deep orange for $\geq 80\%$. Provides an intuitive visual cue without requiring the user to read the exact number.

5. Live Theme Preview with Custom HSV Color Picker

Module: Settings | The colour seed picker shows a live SV gradient area with hue slider and Hex input. The selected colour applies instantly via Material 3's colorSchemeSeed. Users preview changes in real time before committing.

6. Export/Import with Data Integrity

Module: Settings | Export generates a complete JSON snapshot of all user data. Import validates the file format and uses a DB transaction for atomic writes. Errors show a clear message with Retry.

7. Two-Step Clear All Data Protection

Module: Settings | Clearing all data requires two steps: first tap “Clear All Data”, then type “DELETE” in a text field before the destructive button activates. This prevents irreversible data loss from accidental taps.

Annexure B — Screen Wireframes (Calculator, Life Tracker & Settings)

Eight representative screens from the Calculator, Life Tracker, and Settings modules.

